

Generation Interconnection System Impact Study Report

For Queue Project Q22-2053

Cleburne 138 kV

May 25, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2053. The Interconnection Customer proposes a 53.7 MW battery energy storage facility interconnecting at the Cleburne 138 kV substation in the Virginia area. The request was received on April 04, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 442 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,688,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ARCLA SUBSTATION / TWIN LAKES ALTE 138.0 kV	KEARNEYVILLE / CLARK COUNTY 345.0 kV line	106.2	482	30.8
ARCOLA / NEW RD 138.0 kV	CLARK COUNTY / KEARNEYVILLE 345.0 kV L	105.5	453	30.8
NEW ROAD / AG RELIANT 138.0 kV	CLARK COUNTY / KEARNEYVILLE 345.0 kV	105.1	440	30.8
WPCP / TWIN LAKES ALTE 138.0 kV	HNTLEY / TWIN LAKES ALTE 138.0 kV	104.2	273	8.8
AG RELIANT / HUNTINGDON DISTRICT 138.0 kV Line	CLARK COUNTY / KEARNEYVILLE 345.0 kV line	104.0	401	30.8
PASCO CNTY SOLID WASTE RESOURCE RECOVERY / WHEELER 138.0 kV	THOMASTON / LOUDN 345.0 kV line	101.6	649	20.0
JET / HUNTINGDON DISTRICT 138.0 kV L	HUNTINGDON DISTRICT / CLEBURNE SUBSTATION 138.0 kV line	100.8	231	22.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ARCLA SUBSTATION / TWIN LAKES ALTE 138.0 kV	Reconductor line	\$1,212,000	15% (\$179,000)
ARCOLA / NEW RD 138.0 kV	Reconductor line	\$823,000	15% (\$122,000)
NEW ROAD / AG RELIANT 138.0 kV	Reconductor line	\$1,036,000	15% (\$153,000)
WPCP / TWIN LAKES ALTE 138.0 kV	Reconductor line	\$7,897,000	15% (\$1,159,000)
A RELIANT / HUNTINGDON DISTRICT 138.0 kV L	Reconductor line	\$2,590,000	15% (\$383,000)
PASCO CNTY SOLID WASTE RESOURCE RECOVERY / WHEELER 138.0 kV	Reconductor line	\$1,400,000	5% (\$71,000)
JET / HUNTINGDON DISTRICT 138.0 kV	Reconductor line	\$5,304,000	12% (\$621,000)
	Total allocated to project		\$2,688,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.